

CHEMISTRY — SSC-I (Class 9) — Marking Report

Exam-01 · Chapters 1, 2 (Nature of Chemistry, Matter) · Version 1 · Attempted 12 April 2026

Student: Seemab · Date Marked: 12 April 2026 · Total Marks: 65 · Time: 2 h 30 min

11 / 12

SECTION A · MCQS

33 / 33

SECTION B · SHORT

20 / 20

SECTION C · LONG

64 / 65

TOTAL · 98.5% · A+

SECTION A — Multiple Choice Questions (11 / 12)

#	Question (short)	Correct	Seemab	Mark
1	Chemistry deals with composition, structure, properties & changes of	C — Matter	C	1
2	Branch dealing with carbon compounds (except carbonates...)	C — Organic Chemistry	C	1
3	Branch focusing on polymers	B — Polymer	B	1
4	Integration of scientific knowledge for human needs	C — Technology	C	1
5	Forensic chemistry is the application of	B — Analytical	D	0
6	Anything with mass and occupies space	D — Matter	D	1
7	Highly ionized gas, electrons separated from atoms	D — Plasma	D	1
8	BEC occurs at temperatures near	C — $-273.14\text{ }^{\circ}\text{C}$	C	1
9	Diamond, graphite, buckyballs are allotropes of	C — Carbon	C	1
10	Solution with maximum solute at given temperature	D — Saturated	D	1
11	Scattering of light by colloidal particles	C — Tyndall effect	C	1
12	Water is universal solvent because it	B — Dissolves most compounds	B	1

Section A: 11 / 12 (91.7%) — Only one error: confused analytical chemistry with environmental chemistry for forensic applications.

SECTION B — Short Answer Questions (33 / 33)

Q	Topic	Marks	Comment
2(i) OR	Green chemistry & importance	3 / 3	Definition accurate. All three importance points match the key exactly.
2(ii) OR	Geochemistry vs Astrochemistry	3 / 3	Both definitions complete with examples. Earth's crust and astronomical environments clearly distinguished.
2(iii)	Science vs Technology	3 / 3	Definitions, purpose, and solar-energy example all present. Clean tabular layout.
2(iv) OR	Two daily life applications of chemistry branches	3 / 3	Organic (medicines/enzymes) and Physical (batteries/electrochemistry) both correct with applications.
2(v)	Matter & four states	3 / 3	Matter defined; all four states described with examples. Minor: wrote O and H instead of O_2 and H_2 for gas examples — not penalised.
2(vi) OR	Plasma definition & three examples	3 / 3	Ionised gas, electrically conductive. Examples: welding arc, lightning, solar wind. All correct.
2(vii) OR	Compound vs Mixture — three differences	3 / 3	Combination, separation, properties — all three differences clearly stated.
2(viii)	Allotropy & three carbon allotropes	3 / 3	Definition correct. Diamond (tetrahedral, non-conductor), graphite (layers, conductor), buckyballs (C-60, 20 hexagons + 12 pentagons) — all accurate.
2(ix)	Solution, solute, solvent	3 / 3	All three defined with salt-water example. Homogeneous and microscopic particle size noted.
2(x)	Saturated vs unsaturated solutions	3 / 3	Definition, dissolving capacity, and equilibrium all compared. Dynamic equilibrium mentioned for saturated.
2(xi)	Colloid vs suspension	3 / 3	Particle size (1-1000 nm vs $>1000\text{ nm}$), filtration, stability, and examples. Four differences given (three required).

Section B: 33 / 33 (100%) — Perfect score. Every short question answered fully with correct definitions, comparisons, and examples.

SECTION C — Detailed Answer Questions (20 / 20)

Q	Topic	Marks	Comment
3(i)	Five branches of chemistry with definitions & daily life applications	5 / 5	Organic (medicines), Inorganic (Li-ion batteries), Physical (electrochemistry/batteries), Analytical (forensic chemistry), Environmental (water protection). All five complete with definition and application.
3(ii)	States of matter in detail + macroscopic properties	5 / 5	All four states (solid, liquid, gas, plasma) described with properties and examples. Macroscopic comparison table (density, compressibility, fluidity) included. Comprehensive answer.
3(iii)	Elements, compounds, mixtures + classification flowchart	5 / 5	All three defined with examples. Flowchart shows full classification: pure substances → elements/compounds; mixtures → homogeneous (solution → saturated/unsaturated/supersaturated, colloids, suspensions) / heterogeneous. Minor: wrote "Carbondioxide (O ₂)" instead of CO ₂ — not penalised as definition and other examples are correct.
3(iv) OR	Solutions, colloids, suspensions — detailed comparison	5 / 5	All three defined. Detailed comparison table covers: particle type, particle size, filtration, Tyndall effect, stability, and examples. Excellent and thorough.

Section C: 20 / 20 (100%) — All four long answers are comprehensive, well-structured, and accurate. Flowchart in Q.3(iii) is particularly strong.

64 / 65

TOTAL SCORE

98.5 %

PERCENTAGE

A+

GRADE

1

MARKS LOST

Strengths

- **Perfect subjective score** — 53/53 across Section B and C; every definition, comparison, and example is accurate and complete.
- **Excellent comparison tables** — Compounds vs Mixtures, Saturated vs Unsaturated, Colloid vs Suspension, and the three-way Solution/Colloid/Suspension table are all well-structured with correct differentiating points.
- **Strong conceptual understanding of matter** — states of matter, macroscopic properties, allotropy, and classification of matter all handled comprehensively.
- **Branches of chemistry** — all five branches defined with relevant real-world applications (forensic, batteries, medicines, water protection).
- **Flowchart quality** — classification of matter diagram is detailed, showing sub-categories down to saturated/unsaturated/supersaturated solutions.

Areas to Improve

- **Forensic chemistry (MCQ 5):** Forensic chemistry is an application of **analytical chemistry** (examining physical traces, body fluids, drugs), not environmental chemistry. Remember: analytical = identifying unknown substances.
- **Chemical formulas:** Wrote "O" for oxygen gas (should be O₂), "H" for hydrogen (should be H₂), and "Carbondioxide (O₂)" instead of CO₂. Diatomic elements (H₂, O₂, N₂) and molecular formulas matter — examiners may penalise in board exams.
- **Content overlap:** Plasma was described in nearly identical words for both Q.2(v) and Q.2(vi) OR. When answering related questions, try to add new detail rather than repeating the same text.

Exam Results Summary

Exam	Date	Section A	Section B	Section C	Total	%
exam-01-ch1-2 (Nature of Chemistry, Matter)	12 Apr 2026	11 / 12	33 / 33	20 / 20	64 / 65	98.5 %